

Fourier Transform and Weights for Curves

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Abstract

These are notes for the weights seminar in Winter 2025. This follows Mahdi's talk where he discusses how to reduce Deligne's theorem for weights to the case for curves

1 Introduction

We begin by reminding ourselves of some definitions.

- Definition 1.1.** 1. Let $\beta \in \mathbb{R}$. Let $\mathcal{G}_0$ be a Weil sheaf on X_0 and some isomorphism $\tau : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$. Given some closed point $x \in |X_0|$, we set $\bar{x}$ to be a geometric point lying over x . If eigenvalues α of the geometric Frobenius $F_x : \mathcal{G}_{\bar{x}} \rightarrow \mathcal{G}_{\bar{x}}$ satisfy $|\tau(\alpha)|^2 = (\#k(x))^\beta$, then we say that $\mathcal{G}$ is τ -pure of weight β .
2. We say that $\mathcal{G}_0$ is τ -mixed if there is a finite filtration

$$0 = \mathcal{G}^0 \subseteq \mathcal{G}^1 \subseteq \dots \subseteq \mathcal{G}^{r-1} \subseteq \mathcal{G}^r = \mathcal{G}$$

such that $\mathcal{G}^p / \mathcal{G}^{p-1}$ is τ -pure of some weight.

Mahdi previously proved the Riemann hypothesis for finite fields assuming Deligne's theorem, and then he reduced Deligne's theorem to the case for curves.

Recall that Deligne's theorem for curves states the following:

Theorem 1.2. Let $\tau : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ be an isomorphism. Let F be the Frobenius homomorphism, $U = \mathbb{A}_{\overline{\mathbb{F}}_q}^1$, and $\mathcal{G}_0$ a τ -pure lisse $\overline{\mathbb{Q}}_\ell$ -sheaf of weight w . Then, the eigenvalues α of the the Frobenius $f : H_c^1(U, \mathcal{G}) \rightarrow H_c^1(U, \mathcal{G})$ satisfy

$$|\tau(\alpha)|^2 \leq q^{w+1}.$$

To prove Theorem 1.2, we will use Laumon's Fourier transform.

2 The Fourier Transform (but not really)

We are going to talk about a tool (the Fourier transform) which computes cohomology by taking stalks. Specifically, on $\mathbb{A}^1$ what we are going to do is to construct some T_ψ so that

$$T_\psi(\mathcal{G}_0)|_0 = R\Gamma_c(\mathbb{A}_{\overline{\mathbb{F}}_q}^1, \mathcal{G})[1].$$

To do so, we begin by recalling the definition of the Fourier transform for finite fields.

Let $\psi : \mathbb{F}_q \rightarrow \mathbb{C}^\times$ be an additive character. This induces a character on finite extensions $\mathbb{F}_{q^n} \rightarrow \mathbb{F}_q \rightarrow \mathbb{C}^*$ where $\mathbb{F}_{q^n} \rightarrow \mathbb{F}_q$ is the trace map $\text{tr}(\alpha : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_q)$ which we also call ψ . We fix an isomorphism $\tau : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$.

Definition 2.1. The Fourier transform of f is called $T_\psi f : \mathbb{F}_{q^n} \rightarrow \mathbb{C}$ and is defined as

$$(T_\psi f)(x) = \sum_{y \in \mathbb{F}_{q^n}} f(y) \psi(-xy).$$

This satisfies the following properties:

Proposition 2.2. 1. (Plancherel's Formula) Let $\|T_\psi f\|_n^2 = q^n \|f\|_n^2$ where

$$\|g\|_n^2 = \sum_{x \in \mathbb{F}_{q^n}} f(x) \overline{f(x)}.$$

2. (Invertibility) $T_{\psi^{-1}} T_\psi f = q^n f$.

We begin with the “Sheaf-to-functions” correspondence.

Definition 2.3. Given some $K_0 \in D_c^b(\mathbb{A}_0^1, \overline{\mathbb{Q}}_\ell)$, we call $\mathcal{H}^i(K_0)$ to be the sheaf given by taking the sheaf $\ker / \text{Im}$. Then, we define the function

$$f^{K_0} : \mathbb{F}_{q^n} \rightarrow \mathbb{C}$$

given by

$$x \mapsto \sum_i (-1)^i \text{Tr} \left(F_x, \mathcal{H}^i(K_0) \right).$$

We will define a Fourier transform for complexes K_0 called $T_\psi(K_0)$. The functor $T_\psi : D_c^b(\mathbb{A}_0^1, \overline{\mathbb{Q}}_\ell) \rightarrow D_c^b(\mathbb{A}_0^1, \overline{\mathbb{Q}}_\ell)$ is called the functor Fourier transform.

Whatever this is, it will satisfy the following properties:

Theorem 2.4. 1. (Plancherel's Formula) The functor Fourier transform satisfies $\left\| f^{T_\psi(K_0)} \right\|_n = q^{n/2} \|f^{K_0}\|_n$.

2. (Fourier inversion) $T_{\psi^{-1}} T_\psi K_0 = K_0(-1)$.

3. Let $a \in \overline{\mathbb{F}}_q$ be a geometric point of $\mathbb{A}^1 = \mathbb{A}_0^1 \otimes \overline{\mathbb{F}}_q$. Then,

$$(T_\psi(K_0))_a = R\Gamma_c(K \otimes \mathcal{L}(\psi_a))[1]$$

where $\psi_a : \mathbb{F}_{q^n} \rightarrow \mathbb{C}$ is given by $x \mapsto \psi(ax)$ and $\mathbb{F}_{q^n}$ is large enough to contain a .

The point of Theorem 2.4(3) is that to compute cohomology of K , we can compute the stalks of $T_\psi K_0$ instead.

Theorem 2.4. Let X_0 be a curve. We define

$$w(\mathcal{G}) = \sup_{x \in |X_0|} \sup_{\alpha \text{ eigenvalue}} \frac{\log(|\tau(\alpha)^2|)}{\log(\#k(x))}$$

and $w(\mathcal{G}) = -\infty$ when $\mathcal{G} = 0$. Write

$$\|\mathcal{G}\| = \sup \left\{ \rho : \limsup_m \frac{\|f^{\mathcal{G}}\|_m^2}{q^{m(\rho+1)}} > 0 \right\}$$

Then if $H_c^0(\mathbb{A}^1, T_\psi(\mathcal{G})) = 0$ and $T_\psi(\mathcal{G}_0)$ is τ -mixed, then $\|\mathcal{G}\| = w(\mathcal{G})$.

Plancherel's formula can then be restated as $\|T_\psi(K_0)\| = \|K_0\| + 1$.

3 Back to Deligne's Theorem

We begin by proving Theorem 1.2 assuming the following: Let $\mathcal{G}$ be a τ -pure Weil sheaf of weight w , and $\psi : \mathbb{F}_q \rightarrow \overline{\mathbb{Q}}_\ell$ a fixed non-trivial additive character. Then,

1. $T_\psi(\mathcal{G}_0)$ is concentrated in degree 0.
2. $H_c^0(\mathbb{A}^1, T_\psi(\mathcal{G}_0)) = 0$,
3. $T_\psi(\mathcal{G})$ is τ -mixed.

Proof of Deligne's Theorem. We know that to compute cohomology of $\mathcal{G}$, we can compute the stalk of the Fourier transform $(T_\psi(\mathcal{G}_0))$ at a point. Then,

$$(T_\psi(\mathcal{G}_0))|_a = R\Gamma_c(\mathbb{A}^1, G \otimes \mathcal{L}(\psi_0))[1] = R\Gamma_c(\mathbb{A}^1, G)[1].$$

Then by property (1), we know that $T_\psi(\mathcal{G}_0)$ is concentrated in degree 0 so we have that

$$R\Gamma_c(\mathbb{A}^1, G)[1] = H_c^1(\mathbb{A}^1, G)[0].$$

Then, (2) and (3) tell us that the hypotheses of Theorem 2.4 hold. That is, we know that $\|G_0\| = w(G_0)$ and $\|T_\psi(G_0)\| = w(T_\psi(G_0))$. So, to prove Deligne's theorem we need to compute that $\|T_\psi(G_0)\| \leq \|G_0\| + 1$. But this is Plancherel's Formula. $\square$

4 Fourier transforms, for real

4.1 Artin-Schreier sheaves

Let ψ be a character. We consider the Artin-Schreier cover $\wp : \mathbb{A}_0^1 \rightarrow \mathbb{A}_0^1$ given by $x \mapsto x^q - x$. The Galois group of this cover is $\mathbb{F}_q$ so we get a map $\pi_1(\mathbb{A}^1, \bar{x}) \rightarrow \mathbb{F}_q$. Then, composing with the character gives us a map

$$\pi_1(\mathbb{A}^1, \bar{x}) \rightarrow \mathbb{F}_q \rightarrow \mathbb{C}^*.$$

This gives us a rank 1 (étale) sheaf $\mathcal{L}_0(\psi)$. We obtain this sheaf in the following way.

We can view $\mathbb{A}_0^1$ as an $\mathbb{A}_0^1(\mathbb{F}_q)$ -torsor over itself via the Artin-Schreier cover. Then if ψ is a character (i.e., a rank 1 $\overline{\mathbb{Q}}_\ell$ -representation of $\mathbb{A}_0^1(\mathbb{F}_q) = \mathbb{F}_q$, we can obtain a new local system $\mathcal{L}_0(\psi)$. We let $\mathcal{L}_0(\psi) = \mathbb{A}_0^1 \times^{\mathbb{A}_0^1(\mathbb{F}_q)} \psi$.

This funny product is called the contracted product. In the following, we assume both G and H to be abelian so we can avoid worrying about left and right actions. Let X be a G -torsor over some space Z , and suppose we have some group homomorphism $\varphi : G \rightarrow H$. Then we set $X \times H$ to be the torsor on Z with fiber $G \times H$, and we get an action of G on $X \times H$ by acting on the fibers $g \cdot (x, h) = (gx, h\varphi^{-1}(g))$. Then we let $X \times^G H := (X \times H)/G$. This is an H -torsor.

Hence, we get that $\mathcal{L}_0(\psi) = \mathbb{A}_0^1 \times^{\mathbb{A}_0^1(\mathbb{F}_q)} \psi$ is a $\overline{\mathbb{Q}}_\ell$ -torsor on $\mathbb{A}_0^1$. That is, it is a rank 1 $\overline{\mathbb{Q}}_\ell$ -local system on $\mathbb{A}_0^1$.

Example 4.2. To the sheaf $\mathcal{L}_0(\psi)$, the associated function is given by $f^{\mathcal{L}_0(\psi)}(x) = \psi(-x)$. The weight $w(\mathcal{L}_0(\psi)) = 0$.

Proof. Let $\sigma : \overline{\mathbb{F}}_{q^n} \rightarrow \overline{\mathbb{F}}_{q^n}$ be the Frobenius $\sigma(a) = a^{q^n}$.

Then if $a^{q^n} - a = x$, then $\sigma(a) - a = x$. Hence, $\sigma(a) = a + x$. The geometric Frobenius F_x is the inverse of σ , so $F_x(a) = a - x$. As $\mathcal{L}_0(\psi)$ is rank 1, through the character we know that F_x acts on $\mathcal{L}_0(\psi)$ by $\psi(-x)$. So, the trace of the geometric Frobenius F_x is $\psi(-x)$. $\square$

4.2 The Fourier transform

Definition 4.3. Consider the diagram

$$\begin{array}{ccc} & \mathbb{A}_0^1 \times \mathbb{A}_0^1 & \xrightarrow{m(x,y)=xy} \mathbb{A}_0^1 \\ \pi^1(x,y)=x \swarrow & & \searrow \pi^2(x,y)=y \\ \mathbb{A}_0^1 & & \mathbb{A}_0^1 \end{array}.$$

Fix a nonzero character $\psi : \mathbb{F}_q \rightarrow \overline{\mathbb{Q}}_\ell^\times \cong \mathbb{C}^\times$. Then, we define the functor Fourier transform $T_\psi : D_c^b(\mathbb{A}_0^1, \overline{\mathbb{Q}}_\ell) \rightarrow D_c^b(\mathbb{A}_0^1, \overline{\mathbb{Q}}_\ell)$ by

$$T_\psi K_0 = R(\pi^1)_!(K_0 \boxtimes \mathcal{L}_0(\psi))[1].$$

Example 4.4. $f^{T_\psi K_0}(x) = -\sum_{y \in \mathbb{F}_{q^n}} f^{K_0}(y) \psi(-xy)$.

Proof. By definition

$$f^{T_\psi K_0}(x) = \sum_i (-1)^i \text{Tr} \left(F_x, \mathcal{H}^i(R(\pi^1)_!(K_0 \boxtimes \mathcal{L}_0(\psi))[1]) \right).$$

Applying the Grothendieck-Lefschetz trace formula, we get

$$\sum_i (-1)^i \text{Tr} \left(F_x, \mathcal{H}^i(R(\pi^1)_!(K_0 \boxtimes \mathcal{L}_0(\psi))[1]) \right) = \sum_i (-1)^i \sum_{y \in \mathbb{F}_{q^n}} \text{Tr} \left(F_x, \mathcal{H}^i(R(\pi^1)_!(K_0 \boxtimes \mathcal{L}_0(\psi))[1]) \right)$$

$\square$

Theorem 4.5. Let $a \in \overline{\mathbb{F}}_q$ be a geometric point of $\mathbb{A}^1$. Then,

$$(T_\psi(K_0))_a = R\Gamma_c(K \otimes \mathcal{L}(\psi_a))[1]$$

where $\psi_a : \mathbb{F}_{q^n} \rightarrow \mathbb{C}$ is given by $x \mapsto \psi(ax)$ and $n \gg 0$ so that $\mathbb{F}_{q^n}$ contains a .

Proof. This is just base change.

$$(T_\psi(K_0))_a = R\Gamma_c((\pi^2)^* K \otimes m^* \mathcal{L}(\psi)|_{\{a\} \times \mathbb{A}^1})[1] = R\Gamma_c(K \otimes \mathcal{L}(\psi_a))[1].$$

$\square$

5 Proof of the key assertions

Let $j_0 : U_0 \hookrightarrow \mathbb{A}_0^1$ be some open embedding. We now restrict our attention to sheaves of the form $G_0 = (j_0)_!(F_0)$. Here, F_0 is a lisse $\mathbb{Q}_\ell$ -sheaf. Let $\rho : \pi_1(U_0, \bar{a}) \rightarrow V$ be the corresponding representation.

We assume the following, which is the situation Mahdi left us in:

1. F_0 is geometrically irreducible. That is, the associated representation $\rho|_{\pi_1(U, \bar{a})}$ is irreducible.
2. F_0 is geometrically non-constant. That is, the representation $\rho|_{\pi_1(U, \bar{a})}$ is nontrivial.
3. F_0 is unramified at ∞ . That is, ρ factors as

$$\begin{array}{ccc} \pi_1(U_0, \bar{a}) & \xrightarrow{\quad\quad\quad} & \mathrm{GL}(V) \\ & \searrow \quad \quad \nearrow & \\ & \pi_1(U_0 \cup \infty, \bar{a}) & \end{array} .$$

Recall that we want to prove the following key assertions hold:

- (a) The Fourier transform $T_\psi(G_0)$ is concentrated in degree 0.
- (b) $H_c^0(\mathbb{A}^1, T_\psi(G_0)) = 0$.

Proof. (a) Let $x \in \mathbb{A}_0^1(\mathbb{F}_{q^n})$ be a geometric point. Then to check that $T_\psi(G_0)$ is concentrated in degree 0, we just need to check this on stalks. (Is this correct?)

Then, we know by properties of the Fourier transform that

$$(T_\psi G_0)|_x = R\Gamma_c(G_0 \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x))[1].$$

Hence, we want to show that $H_c^0(\mathbb{A}^1, G \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) = H_c^2(\mathbb{A}^1, G \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) = 0$. We are on a curve, so there is of course no higher cohomology.

We note that

$$\begin{aligned} H_c^0(\mathbb{A}^1, G \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) &= H_c^0(U, F \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) \\ H_c^2(\mathbb{A}^1, G \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) &= H_c^2(U, F \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)). \end{aligned}$$

Since F is a lisse sheaf on U and U is affine, we know that $H_c^0(U, F \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) = 0$. So we just need to deal with H_c^2 .

To deal with H_c^2 , we consider the representation $\rho \otimes \psi_x \rightarrow \mathrm{GL}(V)$. By Poincare duality, we know that

$$H_c^{2n}(U, F \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)) = H_c^0(U, F \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)(1))^\vee = (H_c^0(U, F^\vee \otimes_{\overline{\mathbb{Q}}_\ell} \mathcal{L}(\psi_x)^\vee)^\vee(-1).$$

I'm not sure why this isn't immediate (assuming that $F^\vee$ is lisse), since we are taking H_c^0 of a lisse sheaf over U .

- (b) We show that $H_c^0(\mathbb{A}^1, T_\psi(G_0)) = 0$. We have by Fourier inversion that $T_{\psi^{-1}}T_\psi(G_0) = G_0(-1)$. Then,

$$H_c^0(\mathbb{A}^1, T_\psi(G_0)) = R\Gamma_c^{-1}(T_{\psi^{-1}}(T_\psi(G_0)))|_0 = R\Gamma_c^{-1}(G_0(-1))|_0.$$

But $G_0(-1)$ has cohomology concentrated in degree 0, so it has no (-1) -degree cohomology. Therefore, $H_c^0(\mathbb{A}^1, T_\psi(G_0)) = 0$ as desired. □

6 Proof of equality of $\|\mathcal{G}\|$ and $w(\mathcal{G})$

We begin with a definition.

Definition 6.1. Let G_0 be a sheaf on X_0 . Then, the generic maximal weight of G_0 is given by $w_{\mathrm{gen}}(G_0) = w((j_0)^*G_0)$ where $j_0 : U_0 \hookrightarrow X_0$ is any open dense immersion of a smooth U_0 . This quantity is independent of our choice of j_0 .

Theorem 6.2. Let G_0 be a sheaf on X_0 of dimension $\dim X_0 \leq 1$. Let $j_0 : U_0 \hookrightarrow X_0$ be the open subscheme of X_0 consisting of all irreducible components of dimension $\dim X_0$. Then,

1. $\|G_0\| = \max(w_{\text{gen}}((j_0^* G_0)), w(G_0) - 1)$
2. If X_0 is a smooth curve and $H_E^0(X, G)$ for all closed subsets $E \subseteq X$, then $\|G_0\| = w(G_0)$.

Proof. First, we know that if (1) implies (2). This is because if X_0 is smooth of dimension 1, then $U_0 = X_0$ so $w_{\text{gen}}(j_0^* G_0) = w(G_0)$. Hence,

$$\|G_0\| = \max(w_{\text{gen}}((j_0^* G_0)), w(G_0) - 1) = \max(w(G_0), w(G_0) - 1) = w(G_0).$$

So we need to prove (1).

Define $\phi^{G_0}(t) = \sum_{n=1}^{\infty} \|f^{G_0}\|_n^2 t^{n-1}$.

Lemma 6.3. There is a constant C such that

$$\left\| f^{G_0} \right\|_n^2 \leq C \cdot q^{n(w(G_0) + \dim(X_0))}$$

for all $n > 0$. Hence, $\phi^{G_0}(t)$ converges for all $|t| < q^{-(w(G_0) - \dim(X_0))}$.

Proof. Let $x \in |X_0|$ and $d(x) = [k(x) : k]$. By the trace formula,

$$\left| f^{G_0}(x) \right|^2 = \left| \sum_i (-1)^i \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|^2 \leq \sum_i \left| \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|^2.$$

Then, for each $\left| \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|$ we have that

$$\left| \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|^2 \leq r^2 \cdot q^{nw(G_0)}$$

where $r = \max_{x \in X_0} \dim_{\overline{\mathbb{Q}}_\ell}(G_0)_x$ due to the definition of $w(G_0)$ being the largest weight of G_0 and a dimension count.

The r is independent of x , so

$$\begin{aligned} \left\| f^{\mathcal{H}^i(G_0)} \right\|_n^2 &= \sum_{x \in X_0(\mathbb{F}_{q^n})} \left| f^{G_0}(x) \right|^2 \leq \sum_{x \in X_0(\mathbb{F}_{q^n})} \sum_i \left| \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|^2 \\ &\leq \sum_i \sum_{x \in X_0(\mathbb{F}_{q^n})} \left| \text{tr} \left(F_x, \mathcal{H}^i(G_0) \right) \right|^2 \\ &\leq N(\#X_0(\mathbb{F}_{q^n})) \cdot r^2 \cdot q^{nw(G_0)} \leq C \cdot q^{n(w(G_0) + \dim(X_0))} \end{aligned}$$

where N is some number depending on how long the complex G_0 is.

We assume the inequality $N\#X_0(\mathbb{F}_{q^n}) \leq C \cdot q^{n(\dim(X_0))}$ holds. Indeed, for our purposes $X_0 = U_0 \subseteq \mathbb{A}_0^1$ so this is easy. In general, this follows from Noether normalization. $\square$

We note that the radius of convergence of $\phi^{G_0}(t)$ is exactly $q^{-\|G_0\| - \dim X_0}$, and by Lemma 6.3 we have that $\|G_0\| \leq w(G_0)$.

We now can assume that X_0 is connected. This is because

$$\phi^{G_0}(t) = \sum_{W \subseteq X_0 \text{ connected component}} \phi^{G_0|_W}(t)$$

which has radius of convergence the minimum of the radii of the $\phi^{G_0|_W}(t)$. So, $\|G_0\| = \max_W \{\|G_0|_W\|\}$.

6.1 Case 1 - $\dim X_0 = 0$

Let $s \in |X_0|$ be the point. Let $V = (G_0)_s$ be a $\mathbb{C}$ -vector space under $\tau : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$. Then, let $h(t) = \det(I - A \otimes \overline{A} \cdot t^{d(s)})^{-1}$ where $d(s) = [k(s) : \mathbb{F}_q]$.

Then, observe that the logarithmic derivative is given by

$$\sum_{n=1}^{\infty} d(s) \text{tr}((A \otimes \overline{A})^n) (t^{d(s)})^{n-1} = \sum_{n=1}^{\infty} d(s) |\text{tr}(A^n)|^2 (t^{d(s)})^{n-1} = \sum_{n=1}^{\infty} \left\| f^{G_0} \right\|_n^2 t^{n-1} = \phi^{G_0}(t).$$

The radius of convergence is the same as the radius of convergence as $h(t)$ which is the magintude of the smallest pole. Therefore, $\|G_0\| \geq w(G_0)$ since $q^{-w(G_0)}$ the magnitude of a pole of $h(t)$.

6.2 G_0 is a smooth τ -pure sheaf of weight β on a smooth affine curve X_0

Let G_0 be a smooth τ -pure sheaf of weight β on a smooth affine curve X_0 . We assume that X_0 is geometrically irreducible, since we can work over $\overline{\mathbb{F}}_q$.

We let $\overline{G}_0$ be the complex conjugate sheaf $G_0^\vee \otimes \mathcal{L}_{\tau^{-1}(q^\beta)}$ where for some $\alpha \in \overline{\mathbb{Q}}_\ell^\times$ the Weil sheaf $\mathcal{L}_\alpha$ on $\text{Spec } \mathbb{F}_q$ is given by a character of the Weil group $\mathbb{Z} \cong W(k/\mathbb{F}_q) \rightarrow \overline{\mathbb{Q}}_\ell^\times$ where the generator Frobenius is sent to α .

Then, the function

$$L(X_0, G_0 \otimes \overline{G}_0, t) = \prod_{x \in |X_0|} \det\left(1 - F_x t^{d(x)}, (G_0)_{\overline{x}} \otimes (\overline{G}_0)_{\overline{x}}\right)^{-1} = \frac{\det(1 - Ft, H_c^1(X, G \otimes \overline{G}))}{\det(1 - Ft, H_c^2(X, G \otimes \overline{G}))}$$

has logarithmic derivative ϕ^{G_0} . The last inequality uses the trace formula

$$L(X_0, G_0 \otimes \overline{G}_0, t) = \prod_{i=0}^{2 \dim X} \det\left(1 - Ft, H_c^i(X, G \otimes \overline{G})\right)^{(-1)^{i+1}}.$$

We know that $G_0 \otimes \overline{G}_0$ is τ -pure of weight 2β since G_0 is τ -pure of weight β .

1. The radius of convergence of ϕ^{G_0} is at least $q^{-\beta-1}$. This is by the same argument as above (Lemma 6.3).
2. We know that ϕ is the logarithmic derivative of L , so the poles of ϕ^{G_0} are the zeroes and poles of L .

For now, we assume that the poles of are of the form $\frac{1}{\alpha q}$ where α is an eigenvalue of F_x on $(G_0)_{\overline{x}} \otimes (\overline{G}_0)_{\overline{x}}$ for some stalk. Then the assumption on τ -purity,

$$|\tau(\alpha)|^2 = q^{2\beta}.$$

Therefore,

$$\left|\tau\left(\frac{1}{\alpha q}\right)\right| = q^{-\beta-1}.$$

Therefore, any pole of L has norm $q^{-\beta-1}$.

We just need to then justify why the poles of L are of the form $\frac{1}{\alpha q}$. We note that for $j_0 : U_0 \rightarrow X_0$ and $G_0 \otimes \overline{G}_0$, we have that Poincare duality

$$H_c^2(X, G \otimes \overline{G}) = H^0(U, G \otimes \overline{G}^\vee(1))^\vee = ((G \otimes \overline{G})_{\overline{x}}^\vee)^{\pi_1(U, \overline{x})}(-1).$$

Then, α is an eigenvalue of F_x acting on $H_{\overline{x}}$ so F_x on $H_{\overline{x}}^\vee$ has eigenvalue $\frac{1}{\alpha}$. Then, the tate twist lowers gives us $\frac{1}{\alpha q}$ as a pole.

3. For each $x \in |X_0|$, the term

$$L_x = \det\left(1 - F_x t^{d(x)}, (G_0)_{\overline{x}} \otimes (\overline{G}_0)_{\overline{x}}\right)^{-1}$$

is a power series with constant term 1. Note that since $G_0 \neq 0$, each L_x has poles with radius of convergence at least $q^{-w(G_0)} = q^{-\beta}$.

Their poles have norm $q^{-\beta-1}$, so the product of the L_x have radius of converge at most $q^{-\beta-1}$.

Therefore, $\beta = w(G_0) = \|G_0\|$. $\square$